



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Network Security Risk Analyzer – Pre-Attack Assessment

Team ID: DSCPP-III-2026-T131

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026-27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T131
Semester: 3rd
Section: ML-1/2/3
Domain: AI/ML

Team Members

1. Mitushi Jain – 2510017875
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Mentor

Mr. Manish Mehta

Interactions Completed: 1

Problem Statement & Motivation

Problem Statement

Most security teams find out about problems only after something has already gone wrong. Existing scanners can detect vulnerabilities, but they usually report them one by one. They do not clearly show how different weaknesses could connect and form a complete attack path.

As a result, security teams often end up with a long list of alerts without knowing which issues are actually dangerous and should be fixed first.

Motivation

- ▶ Security audits are not always continuous, so new misconfigurations can remain unnoticed for a long time.
- ▶ The longer a weakness remains undetected, the greater the chance of a breach, downtime, or compliance-related problems.
- ▶ Existing scanners often generate too many alerts without helping teams understand which ones should be prioritized.

Target Users / Stakeholders

- ▶ **Primary Users:** Network security engineers, system administrators, and SOC analysts.
- ▶ **Secondary Stakeholders:** CISOs and IT risk managers.
- ▶ **Why it Matters:** The system helps teams understand possible attack paths and focus their time and resources on the risks that matter most.

Project Objectives

1. Build a tool that can automatically scan a network and assess how risky it is.
2. Use graph-based modeling to understand how an attacker could move from one system or service to another.
3. Apply concepts from Data Structures, OOP, Operating Systems, and DBMS in a practical security project.
4. Assign risk scores to vulnerabilities using standard metrics and validate the results with suitable test cases.
5. Create a simple dashboard that clearly shows the risks and helps users decide what should be fixed first.

Key Objective:

Scan the network from an attacker's point of view by checking open ports, exposed services, and weak configurations, then convert the findings into meaningful risk scores and recommendations.

Proposed Approach

How the System Works

1. **Network Discovery:** The system scans the network to identify live hosts, open ports, and running services.
2. **Attack Graph Creation:** The collected information is represented as a directed graph to model possible movement between systems.
3. **Risk Analysis:** CVSS-based calculations are used to estimate the severity of identified vulnerabilities.
4. **Data Storage:** CVE information and previous scan results are stored for future comparison and analysis.
5. **Visualization:** The results are displayed through a dashboard instead of just showing a raw list of alerts.

Key Features

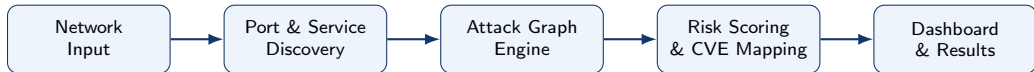
- ▶ Automatically discovers assets and services.
- ▶ Identifies multi-step attack paths.
- ▶ Uses contextual risk scoring instead of only raw CVSS values.
- ▶ Prioritizes vulnerabilities based on their potential impact.
- ▶ Provides recommendations about what should be fixed first.
- ▶ Displays threats through a simple visual dashboard.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Graphs and adjacency lists are used to represent possible attack paths. Hashing helps in faster CVE and port lookups.	Attack Graph Engine
OOPs with C++	Network components are represented using classes. Inheritance and STL containers such as vector and map are used in the core system.	Network Scanner / Core Logic
Operating Systems	Socket programming is used for network probing, while multi-threading helps scan multiple targets without processing them one by one.	Scanning Module
DBMS	SQL and indexing are used to store CVE information, previous scan results, and audit records.	CVE Database / Audit Logs

Integration: The project combines concepts from Data Structures, OOP, Operating Systems, and DBMS to build a complete network risk assessment system.

System Architecture / Workflow



Major Components

- ▶ **Port & Service Discovery Module:** Finds live hosts, open ports, and exposed services.
- ▶ **Attack Graph Engine:** Connects the discovered components and identifies possible paths an attacker could take.
- ▶ **Risk Scoring & Remediation Module:** Calculates risk levels and suggests which vulnerabilities should be addressed first.

Data Flow

- ▶ The scanner first collects information about hosts and services.
- ▶ Vulnerabilities are mapped onto graph nodes to identify possible attack paths.
- ▶ The system calculates risk scores based on the discovered paths.
- ▶ Final results and recommendations are shown on the dashboard.

Technology Stack & Methodology

Technology Stack

- ▶ **Programming:** C++ / C / Python
- ▶ **Database:** SQLite / MySQL
- ▶ **Network Scanning:** Nmap
- ▶ **Graph Visualization:** Graphviz
- ▶ **Development Tools:** VS Code / CLion
- ▶ **Version Control:** Git / GitHub

Development Methodology

1. Understand the project requirements.
2. Design how the scanner, graph engine, and database will connect.
3. Develop the core modules.
4. Test whether vulnerabilities and attack paths are detected correctly.
5. Check performance on larger network environments.

The selected tools make it easier to handle network scanning, vulnerability storage, graph analysis, and result visualization.

Initial Progress & Team Contribution

Progress Achieved

- ▶ Studied existing work related to attack graphs and CVSS scoring.
- ▶ Designed the initial system architecture and planned the data flow between different modules.
- ▶ Developed a basic socket-based network scanner prototype.
- ▶ Prepared an initial database structure for CVE information and vulnerability lookups.

Individual Contribution

Member	Role	Contribution
Mitushi Jain	Lead Developer	Worked on architecture design, team coordination, and documentation
Anushka Jain	Core Developer	Worked on the network scanner and attack graph algorithm
Lipika Latwal	DB / Testing	Designed the database schema, collected vulnerability data, and prepared test cases

Roadmap, Expected Outcomes & References

Roadmap

1. **Phase I:** Project proposal, system design, and an initial prototype.
2. **Phase II:** Develop the core engine and integrate the CVE database.
3. **Phase III:** Build the dashboard and perform final testing and evaluation.

Expected Outcomes

- ▶ A working tool for identifying security risks before an attack occurs.
- ▶ Risk reports that provide priorities instead of just a large number of alerts.
- ▶ A visual representation of the network attack surface.
- ▶ A system that can reasonably handle larger network environments.

References

1. NIST National Vulnerability Database (NVD).
2. CVSS v3.1 Standards.
3. MITRE ATT&CK Framework for Enterprise Networks.
4. Graph Theory Applications in Network Security Modeling.
5. Nmap Security Scanner Documentation.
6. POSIX Socket Programming Guides.

Thank You
Questions & Suggestions